

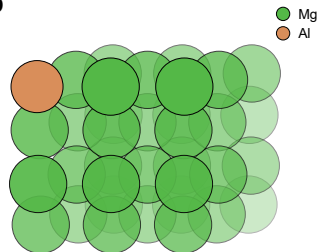
Interface model: Mg(0001)+cation+Al-anion

a



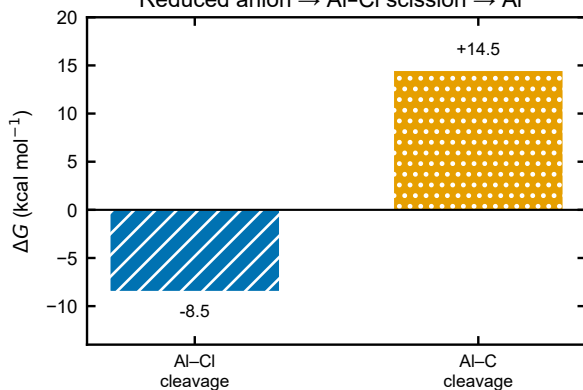
Al in Mg(0001): $\Delta E_{\text{sub}} = -4.44$ eV

b



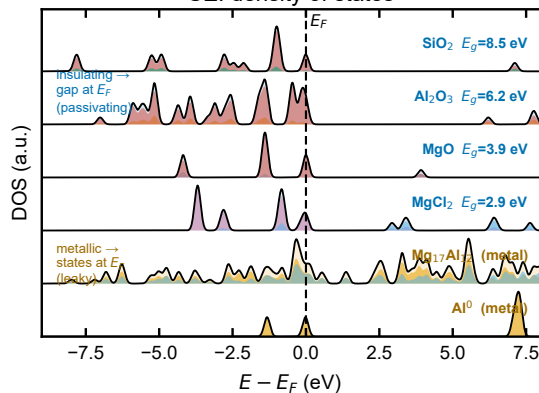
c

Reduced anion $\rightarrow$ Al-Cl scission $\rightarrow$ Al⁰



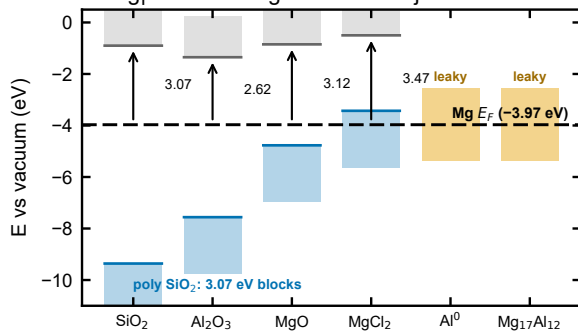
d

SEI density of states



e

Mg|SEI band alignment: e⁻-injection barrier



f

POSS network sequesters Al-anion

